Adaptive Hybrid Control Error for Vehicle Platoons via Reinforcement Learning

1st Report

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Contents

1 Introduction **1**
1.1 A Short Description . . . 1
1.2 Motivation . . . 1
1.3 Our contributions . . . 1

2 Traffic Environment **1**
2.1 Driver Models . . . 1
2.1.1 Intelligent Driver Model (IDM) . . . 1
2.1.2 Stochastic Intelligent Driver Model (SIDM) . . . 2
2.1.3 Connected Autonomous Vehicle (CAV) Model . . . 2
2.2 Mixed Traffic Model . . . 3
2.3 Macroscopic Information . . . 4

3 Reinforcement Learning **4**
3.1 Fundamentals . . . 5
3.2 Reward design . . . 5
3.3 PPO . . . 6
3.4 SAC . . . 6
3.5 Role of RL in the proposed framework . . . 7

4 System Architecture **7**
4.1 Overall structure . . . 7
4.2 Simulation layer . . . 8
4.3 Mesoscopic perception layer . . . 8
4.4 Mesoscopic adaptation layer . . . 8
4.5 Control layer . . . 9

5 Current Progress and Future Work **9**
5.1 Branch overview . . . 9
5.2 Experimental results . . . 10
5.2.1 Branch A results (CPU, 800 episodes) . . . 10
5.2.2 Branch B results (GPU, PPO evaluation) . . . 11
5.3 Comparison between the two branches . . . 11
5.4 Discussion . . . 12
5.5 Future work . . . 12

1 Introduction

1.1 A Short Description

The modernization of intelligent transportation systems (ITS) has necessitated a paradigm shift from individual vehicle autonomy to cooperative automation. Vehicular platooning offers a promising solution to traffic congestion, fuel inefficiency, and road safety.

Thus, our work motivates an "Adaptive Hybrid Control strategy" using RL to adjust the weight between micro and macro. This transition from the standard Adaptive Cruise Control (ACC) to Cooperative Adaptive Cruise Control (CACC) introduces challenges in string stability, the ability to attenuate disturbances, especially under communication delays.

1.2 Motivation

A fundamental trade-off exists in platoon control architectures:

- **Local Control:** Relies on local sensors (radar/LiDAR). It is robust to network failures and ensures collision avoidance but is prone to string instability in long platoons.
- **Global Control:** Relies on V2V data from the leader. It provides excellent string stability but is highly sensitive to communication delays and packet loss.

1.3 Our contributions

- Modeling mixed traffic with human-driven(HDVs) and connected autonomous vehicles (CAVs), improving capacity and stability, in addition to extending traffic analysis from open platoons to closed ring-road systems.
- Proposing a mesoscopic RL framework combining micro- and macro-level traffic states with incorporated resistance forces on our input to create the nearest real life simulation.

2 Traffic Environment

2.1 Driver Models

2.1.1 Intelligent Driver Model (IDM)

The Intelligent Driver Model (IDM) is a deterministic microscopic car-following model describing the longitudinal dynamics of vehicle i as a function of its own velocity, the spacing to its predecessor, and their relative velocity [1]. The acceleration of vehicle i is defined as

$$\dot{v}_i = a_{max}\left[\underbrace{1-\left(\frac{v_i}{v_d}\right)^{\delta}}_{\text{free-road acceleration}} - \underbrace{\left(\frac{s^*(v_i,\Delta v_i)}{s_i}\right)^2}_{\text{interaction (braking) term}}\right]. \tag{1}$$

v_i: velocity of vehicle i,
v_d: the desired free-flow velocity
a_{max}: the maximum acceleration
δ: an acceleration exponent controlling how smoothly the vehicle approaches the desired velocity in free traffic.

When $v_i \ll v_0$, the free-road acceleration $\simeq 1$, so the vehicle accelerates approximately with magnitude a_{max}. As v_i approaches v_0, the term smoothly converges to zero, ensuring bounded acceleration and asymptotic convergence to the desired free-flow equilibrium [1].

The interaction term dominates when the actual spacing s_i is less than the desired spacing s^*, thereby inducing decceleration. The quadratic structure ensures strong braking for small gaps and guarantees collision-free behavior under realistic parameter choices [1].

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{2}$$

where x_i and x_{i-1} denote the longitudinal positions of vehicle i and its leader, respectively, and L is the vehicle length. The relative velocity is defined as $\Delta v_i = v_i - v_{i-1}$.

The desired dynamic spacing is given by

$$s^*(v_i, \Delta v_i) = \underbrace{s_0}_{\text{jam distance}} + \underbrace{v_i T_H}_{\text{time-headway term}} + \underbrace{\frac{v_i \Delta v_i}{2\sqrt{a_{max} b}}}_{\text{anticipation term}} . \tag{3}$$

The constant term s_0 represents the minimum spacing in standing traffic. The linear term $v_i T_H$ reflects the constant time-headway policy characteristic of human driving behavior. The third term, proportional to $v_i \Delta v_i$, introduces anticipative braking by accounting for relative velocity differences and the maximum acceleration and desired deceleration denoted respectively by a_{max} and b [1].

2.1.2 Stochastic Intelligent Driver Model (SIDM)

To account for perception errors, behavioral variability, and unmodeled disturbances, the IDM can be extended by introducing a stochastic acceleration component [2]. The stochastic IDM (SIDM) is formulated as

$$\dot{v}_i = \underbrace{f(s_i, v_i, v_{i-1})}_{\text{deterministic IDM}} + \underbrace{\xi_i(t)}_{\text{stochastic acceleration noise}} , \tag{4}$$

where $f(\cdot)$ denotes the deterministic IDM acceleration function and $\xi_i(t)$ is a stochastic process representing white acceleration noise.

The stochastic term satisfies

$$\langle \xi_i(t) \rangle = 0, \tag{5}$$

$$\langle \xi_i(t) \xi_j(t') \rangle = Q\, \delta_{ij}\, \delta(t - t'), \tag{6}$$

where Q denotes the noise intensity, δ_{ij} is the Kronecker delta ensuring independence between different vehicles, and $\delta(t-t')$ is the Dirac delta distribution representing temporal white noise [2].

Close to the string stability threshold, noise can generate spatiotemporal correlations and traffic oscillations resembling stop-and-go waves. Consequently, the SIDM provides a unified framework for analyzing deterministic instabilities and noise-induced traffic dynamics [2].

2.1.3 Connected Autonomous Vehicle (CAV) Model

For connected autonomous vehicles, we adopt the point-mass longitudinal dynamics described in [3]. The motion of vehicle i is governed by

$$\dot{x}_i = v_i, \qquad \dot{v}_i = u_i, \tag{7}$$

In a car-following configuration, the spacing and relative velocity are defined as

$$s_i = x_{i-1} - x_i - L, \qquad \Delta v_i = v_i - v_{i-1}. \tag{8}$$

CAVs implement an Adaptive Cruise Control (ACC) strategy based on a Constant Time Headway (CTH) policy. The desired spacing is given by

$$s_i^{\text{ref}} = s_0 + T_C v_i, \tag{9}$$

where s_0 denotes the standstill spacing and $T_C > 0$ is the selected time headway.

Defining the spacing error

$$e_{p,i} = s_i - s_i^{\text{ref}}, \tag{10}$$

its dynamics follow from

$$\dot{s}_i = -\Delta v_i, \qquad \dot{s}_i^{\text{ref}} = T_C \dot{v}_i = T_C u_i, \tag{11}$$

which yields

$$\dot{e}_{p,i} = -\Delta v_i - T_C u_i. \tag{12}$$

A linear ACC controller that regulates both spacing and velocity errors can therefore be written as

$$u_i = k_s(s_i - s_i^{\text{ref}}) - k_v \Delta v_i, \tag{13}$$

where $k_s > 0$ and $k_v > 0$ denote the spacing and velocity feedback gains, respectively. This control law constitutes a linear realization of the CTH policy and is consistent with the mesoscopic ACC formulation proposed in [3].

An important property of the CTH-based ACC strategy is its capability to guarantee weak string stability using only local measurements [4]. In the frequency domain, the disturbance propagation transfer function satisfies

$$\|H(j\omega)\|_\infty \leq 1, \tag{14}$$

ensuring that spacing disturbances are not amplified upstream along a vehicle string. Moreover, robustness with respect to sensing and actuation delays imposes a constraint on the admissible implementation lag. In particular, the total delay must satisfy

$$T_C \leq \frac{h_w}{2}, \tag{15}$$

indicating that allowable delay is directly limited by the chosen time headway [4].

2.2 Mixed Traffic Model

We consider a mixed traffic system composed of N vehicles circulating on a single-lane ring road of length L_{ring}. The vehicle indices satisfy periodic boundary conditions such that vehicle 1 follows vehicle N.

Let $x_i(t)$ and $v_i(t)$ denote the position and velocity of vehicle $i \in \{1, \ldots, N\}$. The ring-road constraint is given by

$$x_{i-1}(t) > x_i(t), \qquad x_0(t) := x_N(t) + L_{\text{ring}}. \tag{16}$$

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{17}$$

where periodicity applies for $i = 1$.

Let $\mathcal{H}$ denote the set of HDVs and $\mathcal{C}$ the set of CAVs, with penetration rate

$$P = \frac{|\mathcal{C}|}{N}. \tag{18}$$

Human-driven vehicles are modeled according to the stochastic Intelligent Driver Model introduced in Subsection 2.1.2.

HDVs do not utilize acceleration information of the preceding vehicle, reflecting the absence of communication capabilities. Furthermore, HDVs are characterized by a larger effective reaction time T_H, representing the delay associated with human perception and actuation. Consequently, their response to traffic perturbations is slower and more susceptible to amplification of stochastic fluctuations.

Connected autonomous vehicles are modeled according to the ACC framework introduced previously in 2.1.3. In contrast to HDVs, CAVs are assumed to exhibit higher sensitivity coefficients and a shorter desired time headway T_C, reflecting their faster sensing [5]. The condition $T_C < T_H$ therefore captures the higher responsiveness of CAVs compared to HDVs.

When a CAV follows an HDV, communication-based acceleration information is unavailable. In this degraded configuration, the feedforward gain is set to $k_a = 0$, and the effective time headway is increased to $T_C^{\text{deg}} > T_C$, representing reduced cooperative capability. This modeling choice captures the performance degradation of CAVs in mixed traffic reported in [5] and [6].

The closed-loop mixed system is therefore

$$\dot{v}_i = \begin{cases} f_{\text{IDM}}(s_i, v_i, v_{i-1}) + \xi_i(t), & i \in \mathcal{H}, \\ k_s(s_i - s_i^{\text{ref}}) - k_v \Delta v_i + k_a a_{i-1}, & i \in \mathcal{C}. \end{cases} \tag{19}$$

2.3 Macroscopic Information

Up to this point, only microscopic vehicle interactions have been considered. To improve the performance of the proposed framework, macroscopic traffic information are introduced according to [3]. In particular, we define the mean velocity on the ring as

$$\bar{v}(t) = \frac{1}{N} \sum_{i=1}^{N} v_i(t), \tag{20}$$

and the velocity variance

$$\sigma_v^2(t) = \frac{1}{N} \sum_{i=1}^{N} \left(v_i(t) - \bar{v}(t)\right)^2. \tag{21}$$

The mean velocity represents the macroscopic traffic state, while the variance quantifies the level of speed dispersion and therefore the intensity of oscillations. As discussed in [3], the variance provides a compact indicator of transient traffic behavior and string instability.

3 Reinforcement Learning

In this project, reinforcement learning (RL) is used to improve the behavior of connected and automated vehicles (CAVs) in mixed traffic. The main idea is not to replace the baseline

controller, but to add an adaptive layer on top of it. This learned layer adjusts the time headway policy according to the current traffic situation. In this way, the controller keeps its physical interpretation while gaining the ability to react to complex traffic patterns that are difficult to model explicitly.

3.1 Fundamentals

The RL problem is formulated as a Markov Decision Process (MDP),

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P represents the traffic dynamics, and r is the reward.

At each time step, the agent observes the traffic state, selects an action, applies it to the controller, and receives a reward. In our framework, the state of each CAV i is defined as

$$x_i = (\mu_{v,i}, \sigma^2_{v,i}, v_i, s_i, \Delta v_i, a_{\ell(i)}, \alpha_i^{prev}),$$

where $\mu_{v,i}$ and $\sigma^2_{v,i}$ are the upstream mean velocity and velocity variance, v_i is the vehicle speed, s_i is the spacing to the leader, Δv_i is the relative speed, $a_{\ell(i)}$ is the leader acceleration, and α_i^{prev} is the previously applied headway scaling factor.

This state combines *microscopic* information, such as spacing and relative velocity, with *mesoscopic* information, such as the average speed and speed variance of upstream traffic. This is important because local measurements alone may not be enough to detect the development of traffic waves.

The action is a residual correction of the headway scaling:

$$\mathcal{A} = \{\Delta\alpha_i \in [-\Delta\alpha_{\max}, +\Delta\alpha_{\max}]\}.$$

The final headway scaling factor is then given by

$$\alpha_i(t) = \text{clip}\left(\alpha_i^{rule}(t) + \Delta\alpha_i(t)\right),$$

where $\alpha_i^{rule}(t)$ is the baseline headway scaling provided by the rule-based mesoscopic module. Therefore, RL only refines the baseline decision instead of replacing it. The learned policy can be written as

$$\pi : \mathcal{S} \to \mathcal{A}, \qquad \Delta\alpha_i(t) = \pi(x_i(t)).$$

3.2 Reward design

The reward function is designed to encourage safe, smooth, and efficient traffic behavior while improving string stability. The total reward is

$$r_t = r_{\text{sf}}[t] + r_{\text{sp}}[t] + r_{\text{ef}}[t] + r_{\text{cf}}[t] + r_{\text{ss}}[t] + r_{\sigma^2}[t].$$

The desired spacing is defined by

$$s_i^{des}[t] = d_0 + h_i'[t]\, v_i[t],$$

and the spacing error is

$$e_i[t] = s_i[t] - s_i^{des}[t].$$

The different reward terms have a clear physical meaning:

- r_{sf}: safety term, penalizing unsafe spacing and too small time headway;
- r_{sp}: spacing regulation term, encouraging tracking of the desired spacing;
- r_{ef}: efficiency term, encouraging the vehicle speed to remain close to the upstream mean speed;
- r_{cf}: comfort term, penalizing high jerk;
- r_{ss}: string stability term, penalizing amplification of spacing errors along the platoon;
- r_{σ^2}: traffic smoothness term, penalizing large velocity variance in the traffic stream.

This reward formulation is useful because it combines several objectives into a single learning signal. It allows the RL agent to balance local vehicle behavior and global traffic quality.

3.3 PPO

Proximal Policy Optimization (PPO) is a policy-gradient RL algorithm that updates the policy in a stable and conservative way. Its main advantage is that it is simple to implement and usually robust during training. PPO avoids large policy updates by using a clipped objective:

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right],$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)},$$

and $\hat{A}_t$ is the estimated advantage.

For our problem, PPO is attractive because the action is continuous and the controller must remain stable during learning. It is a good candidate for a first implementation because of its simplicity and reliability.

3.4 SAC

Soft Actor-Critic (SAC) is another RL algorithm for continuous control. Unlike PPO, SAC is an off-policy method and includes an entropy term in its objective:

$$J(\pi) = \sum_t \mathbb{E}\left[r(s_t, a_t) + \beta \mathcal{H}(\pi(\cdot|s_t)) \right],$$

where $\mathcal{H}(\pi(\cdot|s_t))$ is the policy entropy and β controls the exploration level.

The main interest of SAC is its good sample efficiency and its strong performance in continuous-action problems. Since our action is a continuous residual correction $\Delta\alpha_i$, SAC is also well suited to this framework. However, compared with PPO, SAC is usually more complex to tune and explain. We also chose to consider SAC because the mixed-traffic environment is stochastic due to HDV behavior. SAC learns from a replay buffer containing past interactions, which allows it to reuse previous experience instead of relying only on newly collected data. This can make learning more sample-efficient and potentially more robust to traffic variability. In contrast, PPO uses mainly the data generated by the current policy at each training iteration. For this reason, PPO is often easier to implement, while SAC is an interesting alternative for noisy and stochastic environments.

3.5 Role of RL in the proposed framework

The role of RL in our approach is to make the controller adaptive while preserving the original control structure. The baseline controller still ensures a physically meaningful longitudinal behavior, while RL improves the headway adaptation using both microscopic and mesoscopic traffic information.

This integration has three main advantages. First, it keeps the controller interpretable for a control-theory audience. Second, it allows the CAV to react not only to its immediate leader, but also to collective upstream traffic conditions. Third, it avoids solving a complex optimization problem online at each time step. For these reasons, RL is a suitable tool for improving stability and smoothness in mixed traffic.

4 System Architecture

The proposed system is organized as a layered control architecture for mixed traffic on a ring road. Its main objective is to combine microscopic control with mesoscopic traffic awareness. The complete framework can be divided into four main parts: the simulation layer, the mesoscopic perception layer, the mesoscopic adaptation layer, and the control layer. Each part has a specific role, and together they allow the CAVs to react not only to their immediate leader but also to the collective traffic conditions upstream.

4.1 Overall structure

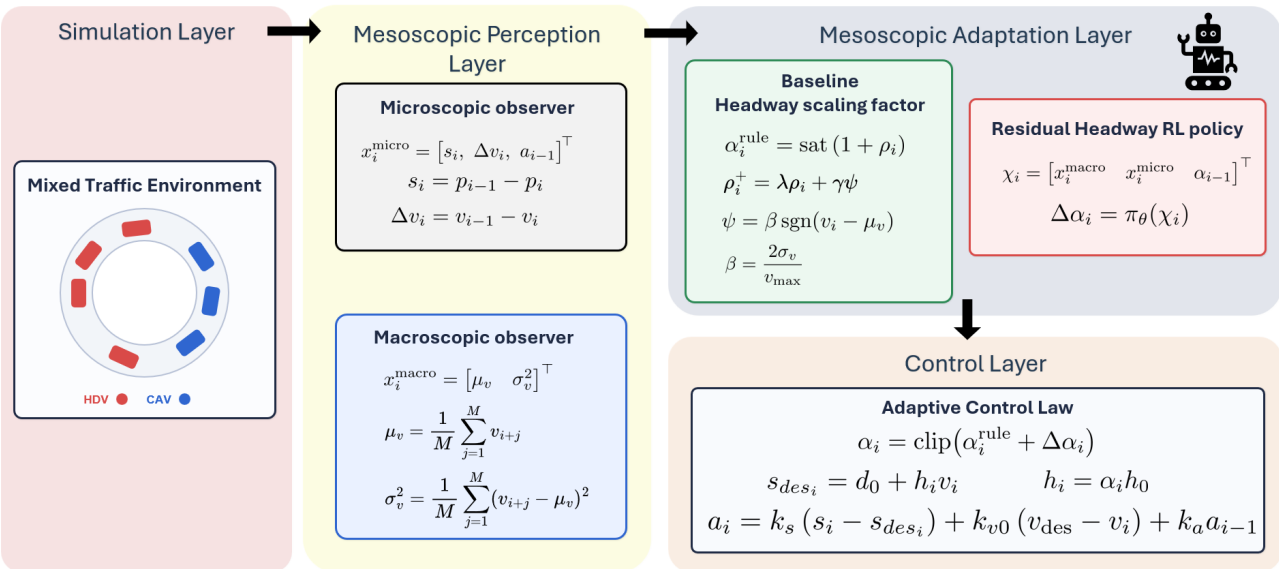

Figure 1: Overview of the proposed system architecture.

The simulation layer represents the mixed-traffic environment, where human-driven vehicles (HDVs) and connected and automated vehicles (CAVs) evolve on the same road. At each time step, the CAV receives observations from the traffic environment, processes them through two perception channels, computes an adaptive parameter, and finally applies the resulting control law. In this way, the architecture separates *observation*, *adaptation*, and *actuation*, which makes the framework easier to interpret from a control perspective.

4.2 Simulation layer

The simulation layer contains the traffic environment and generates the variables needed by the control framework. The road is modeled as a ring road, which avoids boundary effects and makes the appearance of traffic oscillations easier to study. HDVs and CAVs coexist in the same environment, but only the CAVs use the adaptive control architecture.

This layer provides the real-time traffic variables used by the upper layers, such as positions, velocities, spacing, and acceleration. It is therefore the source of both local and collective traffic information.

4.3 Mesoscopic perception layer

The perception layer extracts the information used by the controller. It contains two complementary observers.

Microscopic observer. The microscopic observer describes the local car-following situation of vehicle i. It uses the spacing, relative velocity, and ego speed:

$$x_i^{\text{micro}} = \begin{bmatrix} s_i & \Delta v_i & a_{i-1} \end{bmatrix}^\top,$$

with

$$s_i = p_{i-1} - p_i, \qquad \Delta v_i = v_{i-1} - v_i.$$

These variables describe the immediate interaction with the preceding vehicle and are the standard quantities used by longitudinal controllers.

Macroscopic observer. The second observer captures collective traffic behavior upstream of the CAV. Instead of looking only at the leader, it summarizes the traffic state over a perception horizon. The two main quantities are the upstream mean velocity and the upstream velocity variance:

$$x_i^{\text{macro}} = \begin{bmatrix} \mu_v & \sigma_v^2 \end{bmatrix}^\top,$$

with

$$\mu_v = \frac{1}{M}\sum_{j=1}^{M} v_{i+j}, \qquad \sigma_v^2 = \frac{1}{M}\sum_{j=1}^{M}(v_{i+j} - \mu_v)^2.$$

The mean velocity gives an estimate of the average upstream traffic speed, while the variance indicates how irregular or unstable the traffic stream is.

4.4 Mesoscopic adaptation layer

The adaptation layer transforms the observed traffic information into a headway modification for the CAV controller. It contains two parts: a baseline adaptive headway mechanism and a residual reinforcement learning policy.

Baseline adaptive headway. The baseline mechanism computes a rule-based headway scaling factor from the macroscopic traffic information. The update can be written as

$$\alpha_i^{\text{rule}} = \text{sat}(1 + \rho_i),$$

with the internal state

$$\rho_i^+ = \lambda \rho_i + \gamma \psi,$$

and the perception signal

$$\psi = \beta \,\mathrm{sgn}(v_i - \mu_v), \qquad \beta = \frac{2\sigma_v}{v_{\max}}.$$

This mechanism has a simple interpretation. When upstream traffic becomes more variable, the controller increases the effective headway in order to react more cautiously. When traffic is smoother, the headway can be reduced. The filtering through ρ_i also avoids abrupt parameter changes.

Residual RL policy. On top of the baseline adaptation, a residual policy computes a correction term:

$$\chi_i = \begin{bmatrix} x_i^{\text{macro}} & x_i^{\text{micro}} & \alpha_{i-1} \end{bmatrix}^\top, \qquad \Delta\alpha_i = \pi_\theta(\chi_i).$$

The role of this block is to refine the rule-based decision without changing the overall structure of the controller. In other words, the adaptive law remains physically interpretable, while the learning block adds flexibility when the traffic situation becomes more complex.

4.5 Control layer

The control layer converts the adapted headway into a longitudinal acceleration command for the CAV. The final scaling factor is

$$\alpha_i = \mathrm{clip}\big(\alpha_i^{\text{rule}} + \Delta\alpha_i\big)\,,$$

and the corresponding time headway becomes

$$T_i = \alpha_i T_0.$$

This adapted headway is then used inside the microscopic CACC/CTH controller:

$$a_i = k_s\,(s_i - d_0 - T_i v_i) + k_v \Delta v_i + k_a a_{i-1} + k_{v0}(v_{\text{des}} - v_i).$$

This equation shows clearly how the architecture is organized. The lower-level controller remains a classical feedback law based on spacing, relative speed, leader acceleration, and desired speed regulation. The upper layers do not replace this controller; they only adapt the time-headway parameter T_i used inside it.

5 Current Progress and Future Work

Two parallel development branches have been pursued in this project. Each branch implements the same core framework described in the previous sections, namely a residual RL policy on top of a mesoscopic adaptive headway rule, but the two branches differ in their engineering choices and experimental focus. In this section, we summarize the current state of each branch, compare the experimental results obtained so far, and outline the remaining work.

5.1 Branch overview

Branch A (sequential CPU implementation). This branch follows a lightweight, single-episode training loop running entirely on CPU. The RL module is embedded inside the mesoscopic adaptation layer: the rule-based adapter first computes a headway scaling factor α^{rule}, and the RL policy then outputs a bounded residual correction $\Delta\alpha \in [-0.1, 0.1]$. The observation is an 8-dimensional vector that captures both local car-following variables and upstream mesoscopic statistics:

$$x_i = \big(\mu_v,\ \sigma_v^2,\ \delta_v,\ v_i,\ s_i,\ \Delta v_i,\ a_{\ell(i)},\ \alpha_i^{\text{prev}}\big).$$

Training collects one full episode (1500 time steps) before each policy update. Both PPO and SAC have been implemented and compared over 800 training episodes.

Branch B (vectorized GPU implementation). This branch rewrites the entire ring-road simulator as a set of parallel PyTorch tensor operations, eliminating all Python loops over vehicles. Multiple independent environments (16 to 128) are batched on the GPU, so that a single `step()` call advances all simulations simultaneously. The RL layer is decoupled from the mesoscopic module and placed in its own `agents` package. The observation space is extended to 15 dimensions by adding global context features such as fleet mean speed, CAV penetration rate, and neighbourhood statistics. GPU-native PPO and SAC trainers have been implemented, together with a formal evaluation protocol that sweeps over four control modes (baseline, adaptive, PPO, SAC), four human-driver ratios, two CAV layout strategies (ordered vs. shuffled), and optional perturbation injection.

5.2 Experimental results

Both branches evaluate three control modes, no RL (baseline adaptive headway), PPO, and SAC, across several human-driver penetration rates on the same 22-vehicle, 300 m ring road. The key metrics are mean fleet speed, speed variance, minimum inter-vehicle gap, collision count, and ride comfort (RMS acceleration and jerk).

5.2.1 Branch A results (CPU, 800 episodes)

Table 1 summarizes the evaluation results for Branch A. Each row corresponds to one control mode evaluated at a given human-driver ratio. The most critical scenario is the 75 % human-driver case, where the traffic is dominated by stochastic HDV behavior.

Table 1: Branch A evaluation metrics across human-driver ratios (800 episodes, CPU training).

HR	Mode	Mean speed (m/s)	Speed var. (m^2/s^2)	RMS acc. (m/s^2)	RMS jerk (m/s^3)	Min gap (m)	Collisions
25 %	No RL	4.868	0.630	0.578	8.682	0.510	0
	PPO	**5.081**	0.780	**0.441**	**6.302**	0.510	0
	SAC	4.822	**0.630**	0.746	11.363	0.510	0
50 %	No RL	3.745	1.939	**0.437**	**5.169**	0.510	0
	PPO	3.727	**1.819**	0.439	5.257	0.510	0
	SAC	**3.755**	1.916	0.477	6.011	0.510	0
75 %	No RL	2.982	3.793	0.420	**3.451**	0.307	42
	PPO	**2.992**	**3.608**	**0.419**	3.520	**0.489**	**1**
	SAC	2.977	3.829	0.426	3.591	0.300	42

At 25 % human drivers, all three methods perform well and no collisions occur. PPO achieves the highest mean speed (5.08 m/s) and the best comfort metrics, while SAC matches the baseline variance. At 50 %, the three modes are comparable, with PPO slightly reducing speed variance.

The most significant result appears at 75 % human drivers. The baseline and SAC both produce 42 collisions, whereas PPO reduces this to a single collision, a 98 % improvement. PPO also achieves the best speed variance (3.608 vs. 3.793) and the largest minimum gap (0.489 m vs. 0.307 m). This suggests that the on-policy nature of PPO is better suited to the high-stochasticity regime of this scenario, where HDV behavior dominates.

5.2.2 Branch B results (GPU, PPO evaluation)

Branch B evaluates five training checkpoints across the same human-driver ratios. Additionally, it introduces a perturbation-injection protocol and a formal string-stability metric (the amplification ratio: the ratio of the maximum downstream speed perturbation to the upstream perturbation). Table 2 summarizes the results for the final PPO checkpoint compared with the adaptive and baseline modes.

Table 2: Branch B evaluation metrics (GPU PPO final checkpoint, overall summary).

Mode	HR	Mean speed (m/s)	Speed var. (m^2/s^2)	Min gap (m)	Safety rate (%)	Amplification ratio
Adaptive	0 %	4.829	0.321	7.32	100	0.525
	25 %	4.975	0.011	—	100	—
	50 %	4.587	0.232	—	—	—
Baseline	0 %	5.019	0.040	7.41	100	0.368
	25 %	4.975	0.011	—	100	—
	50 %	4.330	1.229	—	—	—
RL (PPO)	0 %	**7.081**	0.094	6.91	100	0.744
	25 %	**6.865**	0.138	—	—	—
	50 %	4.499	1.145	—	—	—

The GPU-trained PPO agent achieves a substantial speed improvement in the low human-driver regime: +46.5 % over adaptive at 0 % humans and +38.1 % at 25 % humans. However, this comes at the cost of reduced string stability (amplification ratio 0.744, closer to the instability threshold of 1.0). The checkpoint ranking analysis reveals an inherent speed–safety tradeoff: later checkpoints produce higher speeds but progressively worse safety metrics. The best safety–performance tradeoff is found at an early checkpoint (step 229 376), which achieves a 50 % safety pass rate across all evaluated configurations.

5.3 Comparison between the two branches

Although the two branches share the same underlying framework, several design differences lead to complementary strengths and weaknesses.

Observation space. Branch A uses 8 dimensions (local state only), while Branch B uses 15 dimensions (local + global context). The richer observation may help the policy generalize across different penetration rates, but it also increases the number of hyperparameters and makes the learned policy harder to interpret.

RL–mesoscopic coupling. In Branch A, the RL policy operates on top of the mesoscopic adapter, which remains active at runtime and provides a stable baseline. In Branch B, the RL agent is decoupled from the mesoscopic layer. This means that Branch A always benefits from the rule-based fallback, while Branch B gives the RL agent more freedom but also more responsibility.

Training scale. Branch A trains from single-episode rollouts on CPU (approximately 8 hours for 800 episodes). Branch B parallelizes across 16–128 environments per GPU step, enabling over an order of magnitude more transitions per wall-clock second. This allows Branch B to explore a much larger portion of the state space, which may explain the aggressive speed gains observed at low human-driver ratios.

Safety formalization. Branch A monitors collisions and gap clamps as logged quantities. Branch B introduces a formal safety-first checkpoint ranking: checkpoints are first filtered by worst-case collision count, then by minimum gap, and finally by traffic performance. This ensures that no checkpoint is selected purely on speed if it violates safety constraints.

String stability. Branch A uses a binary validity criterion (no gap clamps during the episode). Branch B computes a continuous amplification ratio from a controlled perturbation experiment. The continuous metric is more informative for evaluating the degree of string stability across conditions.

5.4 Discussion

The results so far show that both PPO and SAC can improve traffic behavior compared to the rule-based adaptive baseline, but their advantages depend on the traffic regime and the evaluation metric.

In the high human-driver regime (75 %), Branch A's PPO achieves a dramatic safety improvement (42 collisions reduced to 1), suggesting that the coupled RL–mesoscopic design and the conservative observation space are well suited when HDV behavior dominates. In the low human-driver regime (0–25 %), Branch B's GPU-trained PPO achieves much higher fleet speeds (+38–46 %), likely because the larger observation space and more extensive training allow the policy to exploit the coordinated nature of the all-CAV or mostly-CAV traffic.

An important finding from Branch B is the speed–safety tradeoff across training checkpoints. As training progresses, the agent learns to achieve higher speeds but at the cost of smaller gaps and higher perturbation amplification. This suggests that future reward designs should include explicit safety constraints rather than relying on soft penalty terms alone.

SAC performs comparably to PPO in moderate traffic conditions (25–50 % human drivers) in both branches but struggles at 75 % in Branch A. This may indicate that off-policy learning, while more sample-efficient in theory, is more sensitive to the distribution shift caused by highly stochastic HDV behavior in the single-episode training setting.

5.5 Future work

Unified evaluation. The most immediate next step is to evaluate both branches under an identical protocol: the same ring-road configuration, the same human-driver ratios, the same random seeds, and the same perturbation injections. This will allow a fair comparison between the two observation spaces, reward designs, and training scales.

Reward shaping and safety constraints. The speed–safety tradeoff observed in Branch B motivates the introduction of constrained optimization techniques, such as Lagrangian relaxation or a safety critic, to enforce hard safety bounds during training rather than relying on post-hoc checkpoint filtering.

Merging the best of both branches. A natural synthesis would combine Branch A's coupled RL–mesoscopic control stack (where the rule-based adapter remains active) with Branch B's vectorized GPU simulator and richer observation space. This would retain the safety benefits of the layered architecture while enabling large-scale parallel training.

Extended traffic scenarios. The current evaluation is limited to a single ring-road configuration with 22 vehicles. Future experiments should vary the road length, vehicle count, and

HDV aggressiveness distributions to assess how well the learned policies generalize beyond the training conditions.

SAC improvements. The poor collision performance of SAC at 75 % human drivers deserves further investigation. Possible directions include tuning the entropy temperature schedule, increasing the replay buffer size, or pre-training SAC from a PPO checkpoint (a capability already implemented in Branch B).

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